Final CIA Capstone Report: Homebound

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***Abstract*—Users will play as a child tasked with unpacking their bedroom after moving. We want to both stimulate memory recall and provide a sense of closure for dementia patients. Rehoming and moving can be a very scary and intense process for individuals with dementia. We created an environment that dementia patients could call their own. We place players in a childhood bedroom, populated with toys, objects, and sounds that they would have grown up with in that era (in this case, the 50s and 60s, when users would have been around 12 years old). The player is then tasked with unpacking their room, guided by a motherly figure with spoken dialogue. The toys and items being unpacked are interactive and designed to invoke memories and positive emotions in users.**

I. Introduction

Fig. 1. Ainsley's environment mockup with materials.

A. Motivation

The aging population faces significant challenges when it comes to cognitive decline, particularly in the form of dementia-related memory loss, disorientation, and emotional distress. For many individuals with dementia, relocating to assisted living or memory care facilities can be especially traumatic, as familiar surroundings and routines are often central to their remaining cognitive anchors. While there has been promising research into the use of virtual reality (VR) to support memory recall and reduce anxiety in dementia patients, existing experiences are often generic, task-based, or under-stimulating in terms of personal relevance.

Homebound addresses these shortcomings by offering a guided, emotionally driven VR experience that places users in a simulated childhood bedroom, designed to reflect the 1950s-1960s time period - the era in which most current dementia patients would have been grown up in. The user is gently prompted to explore and pack objects from the environment while being guided by a maternal voice, designed to foster emotional comfort and memory engagement. The interactive environment aims to stimulate memory recall, promote a positive mood, and engage fine motor skills through low-stress activities like coloring, playing a xylophone, and selecting records.

The emotional goal of the experience is to evoke a sense of calm, familiarity, and meaningful closure. By recreating spaces and objects from a user's past and pairing them with gentle guidance and accessible controls, the experience hopes to reduce anxiety and support therapeutic care goals. The project was motivated not only by academic interest but also by a growing unmet need in both commercial and therapeutic VR applications for older adults with cognitive impairments.

B. Summary

Homebound is a VR experience designed to support memory recall and emotional comfort in individuals experiencing dementia. Developed in Unity for the Meta Quest 2 headset, the project simulates a 1950s-1960s childhood bedroom and invites users to interact with familiar objects from that era. The experience is guided by gentle narration and includes three primary modes of interaction: coloring with virtual crayons, playing a functional xylophone, and selecting records to play on a working record player. Each interaction is designed to promote engagement, fine motor coordination, and emotional connection.

The development process emphasized accessibility, comfort, and emotional resonance. Controls were simplified to suit seated users and reduce cognitive load, and aesthetic decisions were made to reflect the sensory preferences and limitations of individuals with dementia. A warm, maternal voice provides soft guidance to foster a sense of security and reduce disorientation.

Validation was conducted through four rounds of user playtesting, with detailed survey responses as feedback for the project. Across these iterations, participant feedback indicated significant improvements in both interaction design and historical immersion. In the earliest round, most participants perceived the room as a different decade than intended, but by the final round, most participants correctly identified the intended time period. Users described the environment as comforting, the controls as increasingly intuitive, and the experience as emotionally impactful. These findings support the project's core goals and suggest further potential for VR in memory care applications.

II. Background / Related Work

C. Similar Works and Competitive Analysis

In recent years, VR has emerged as a promising tool for promoting emotional well-being and cognitive stimulation in individuals with dementia. A number of projects have explored immersive environments as therapeutic interventions, often with a focus on replicating everyday tasks or calming nature scenes. For example, one research study developed a virtual kitchen where users were asked to complete basic household tasks such as setting the table. These simulations provided evidence that VR can improve engagement and recall; however, they often prioritized structured activity over emotional resonance or personal familiarity [4].

Unlike these task-driven environments, *Homebound* focuses on unstructured, emotionally driven exploration. Rather than guiding users through a list of actions, the experience places them in a nostalgic sandbox designed to evoke a personal and emotional response. This distinction is critical in dementia care, where mood regulation and personal connection often have greater therapeutic value than task completion alone.

Several reviews and studies have noted the limited number of VR applications that are specifically tailored for dementia patients. Many existing systems lack the interactivity and environmental specificity needed to truly resonate with older users. Homebound addresses this by creating an immersive setting that recreates the sensory and emotional landscape of a user's childhood, including era-appropriate furniture, toys, sounds, and music [1][5].

Furthermore, *Homebound* places a strong emphasis on multisensory engagement. Auditory feedback from a maternal voice helps reduce anxiety and disorientation, while tactile interactions, such as coloring or playing the xylophone, encourage motor engagement and agency. This approach contrasts with many passive VR environments, which are explorable but lack interactivity.

Lastly, the sandbox structure of the project offers players the freedom to explore and engage with the space at their own pace. This design choice supports autonomy and reduces the risk of cognitive overload, aligning with the best practices for accessibility and emotional safety in dementia-oriented applications.

D. Aesthetic Inspiration

The aesthetic design of *Homebound* was guided by a blend of nostalgic familiarity and inclusivity. The room was styled to resemble a modest mid-century childhood bedroom, drawing inspiration from cabin-like interiors common in suburban American homes during the 1950s through the 1960s. This included warm wood textures, patterned wallpaper, and vintage household items designed to evoke a sense of comfort and timelessness.

To ensure accessibility and relatability for a wide range of users, the environment avoided gendered visual cues. Toys and decorations were selected to be neutral and inviting, reinforcing a space that any user could imagine as their own childhood room. Stuffed animals, building blocks, art tools, and posters were curated to reflect the kinds of belongings commonly found during the target era, emphasizing emotional resonance over realism.

Every object was chosen to feel lived-in and emotionally charged, helping ground the user in a memory-inspired space while also promoting exploration and comfort. The goal was to create an environment that felt both specific in time and universal in emotional tone.

Fig. 2. Aesthetic Inspiration 1

Fig. 3. Aesthetic Inspiration 2

III. Design

The primary goal of *Homebound* was to create an emotionally supportive and accessible VR experience that evokes calm and stimulates memory recall in users with dementia. The design process prioritized comfort, familiarity, and inclusivity while maintaining an engaging and interactive environment that encourages exploration.

E. Emotional and Cognitive Goals

The experience is framed around the act of unpacking a childhood bedroom—an activity that is both simple and emotionally rich. By grounding the player in an era-specific space and incorporating familiar sounds, visuals, and objects, the design seeks to activate personal memories and foster a sense of emotional closure. The use of a maternal voice to guide the experience supports this goal, providing gentle structure and emotional warmth.

F. Core Design Principles

a. Accessibility

The experience was built to be fully playable while seated. Movement is controlled with the left joystick, rotation with the right, and the only required input is the bottom trigger for grabbing objects. Visual cues are displayed on the controllers themselves, and in-headset hand animations reinforce actions for clarity.

b. Reduced Stimulation

To accommodate sensory processing challenges common among dementia patients, the environment avoids visual clutter, uses warm lighting, and features consistent object scale and spatial layout. Color contrast and shadows were carefully tuned to support depth perception and object recognition.

c. Emotional Safety

Instead of high-stakes or goal-oriented tasks, the experience encourages relaxed exploration. There are no time limits or progress meters; users are free to interact with objects or simply observe their surroundings.

d. Narrative Comfort

A maternal voice provides context and guidance through soft-spoken narration. Research suggests maternal tones are often perceived as more calming and nurturing, making this a deliberate choice to reduce stress and support emotional grounding [4].

G. Key Features and Interaction Design

e. Crayons and Easel

Users can pick up vintage-styled crayons and draw on a canvas. The drawing system is built using raycasting and collision detection to apply color to a texture in real-time. Rotation locking, generous collider sizing, and feedback smoothing were all implemented to make this interaction more intuitive.

f. Xylophone and Mallets

Each mallet can be picked up and used to strike keys on a xylophone, each of which plays a distinct pitch. This promotes fine motor skill usage and offers satisfying audio feedback. The system uses physical collisions rather than triggers for a more tactile feel.

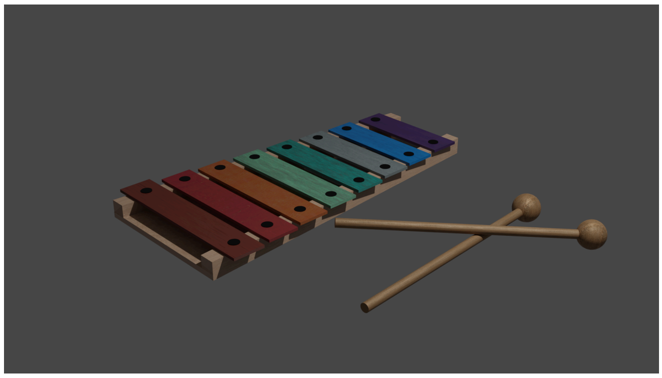

Fig 4. Blender model of xylophone and mallets.

g. Record Player

Users can select and place vinyl records onto a turntable, each corresponding to a different period-appropriate song. The record rotates during playback, and collisions are used to trigger audio in response to physical placement.

H. Differentiation from Related Work

While some VR applications for dementia focus on guided tasks or static environments, *Homebound* sets itself apart through its open-ended, exploratory design. Rather than pushing users toward specific objectives, the environment encourages organic interaction and emotional reflection. It is not a puzzle, nor is it a simulation of future living—it is a curated memory space.

The project also places significant emphasis on authentic sensory immersion. Visual elements were modeled to match the textures, forms, and objects of the 1950s–60s. Sound design included vinyl record audio, subtle ambient noise, and spoken dialogue. The sandbox structure offers a sense of control and freedom, important qualities in therapy-oriented VR design.

The overall feel was informed by game design principles such as "game feel" (Swink) and user-centered emotional design. Rather than designing purely for utility or realism, the experience was built around how it should feel to be in that room, as if stepping into a memory, where emotional truth matters more than mechanical complexity.

Fig. 4. Final design of room. View from bed (top). View from door (bottom)

IV. Implementation

In our ideation phase, we collectively chose to use VR for our project. The initial technology that we had access to was the HTC Vive Cosmos. This proved to be, although performant, bulky and not portable. Beginning in Spring Quarter 2025 we had access to the Meta Quest 2. This proved, because of its portability, to be much easier to work with and demo in class.

We decided to use Unity as our game engine. Godot, although impressive with its growing capabilities, does not have well-developed VR tools and libraries. Unreal, while very powerful, is commonly known for its steeper learning curve that would not be ideal for a twenty week project.

We looked into Unity Version Control, but concluded from the consensus by peers and online forums that it would be best to use GitHub. We were also most familiar with GitHub for version control.

We found and used Unity's XR interactive toolkit as the skeleton for our project. It provided the basic functionality for grabbing and movement. In Winter Quarter, we encountered much difficulty with available online tutorials, and by Spring Quarter, our primary support for using this library was the default scene generated by Unity's VR template.

I. XR Grab Interactables

XR grab interactables were our most commonly used component from the XR library. The primary configuration we altered was the default far attach mode. This meant that if a player pointed at an object, they had to use the joystick to physically pull the object close to them. This was very unintuitive, as we saw from our first playtest, and after exploring Unity's settings for XR Grab interactables, we set the default attach mode to "near."

J. Crayons and Easel

Claire started scripting with the Crayon and Easel assets. This part of the project focused largely on understanding Unity's XR grab interactable script and inspector component and box colliders.

We began by using a ray casting algorithm drawn from marker assets to apply a texture to a whiteboard [3]. Following the tutorial, they created a prefab for the crayon, with three components, a tip, body, and grab point. The tip of the marker was used for the raycast and had the material with the color to sample from. The body of the marker had the XR grab interactable component, and the grab point was a transform applied to the attach transform. This algorithm drew smooth lines, however, as one responder indicated from our first playtest, "the weirdest thing was you drew onto the paper a good like 5 cm or so away from the paper" [6]. In other words, the simple raycast was somewhat unintuitive and made it difficult to "remove" the marker from the surface of the whiteboard.

For the next phase of the project, Ainsley modeled some vintage crayon assets to replace the markers, Emily modeled the easel asset to replace the whiteboard, and Claire adapted the script accordingly. The main marker/crayon prefab remained the same, but the script had to be modified according to the feedback from the playtest.

```
init_whiteboard()
init_markers()

MARKER:
on collision with whiteboard:
   add contact point to queue
   update last draw time
on update:
   lock pen rotation
   process points of contact in queue
       lerp between points

WHITEBOARD:
if (timeElapsed and changesMade):
   applyTexture(whiteboard)
```

Fig. 5. Crayon/easel script pseudocode.

We locked rotation upon contact to the easel, so that the crayons would not strangely interact with the easel. Using a combination of a raycast and a trigger collision, we were able to both detect if the crayon was intersecting with the whiteboard and also get a precise pixel coordinate in the easel texture to draw on. We learned that the trigger collisions were occurring multiple times per frame, so we developed a system to limit the number of points drawn per frame to avoid the performance issues that arose. We also enlarged the collider boxes on the crayon tips to provide a more generous coloring allowance.

One recurring issue that was tedious to deal with was the grab point attach transform applied to the crayons. The attach transform was unintuitive to modify as the transforms were relative to the object and not in world space.

Playtests showed that over time, the coloring mechanic became more intuitive. In round three of playtesting, responders had the following to say: “the crayons didn’t draw too accurately”; “I had to go pretty close to the canvas which made it a bit hard”; “sometimes I felt like I was close enough to draw but I wasn't”; “it was a tiny bit tedious.” The final round of playtesting had more positive responses: a lot smoother and better than last time, “the angle on the crayons are good,” and “the crayons were easy to use and to pick up and switch between colors” [6].

In the future, we intend to refine the angle that the crayon is held and the smoothness of the drawing to create a more comfortable and engaging experience.

K. Xylophone

We primarily used the Unity documentation for information about AudioSource objects [7].

Because we wanted every key to have a different pitch, we created a very simple script attached to each xylophone key that on a physical collision with a mallet tagged object, would play the audio clip from that key’s AudioSource component.

The biggest issue we encountered with the xylophone interactivity was configuring the grab position for the mallets, understanding trigger vs physical collisions, and general settings for xr grab intractable objects. We decided to use physical collisions for this because trigger collisions caused the mallets to phase through the xylophone.

Earlier playtests indicated that the collision for the mallets was too imprecise, and in response to this, we reduced the size of the collision boxes [6].

L. Record Player

Each record’s object name corresponds to an index into a song list attached to the record player. The record player first checks for a physical collision with a “Record” tagged object, then checks the object name for the specific song index. Before playing, the record object is translated to the record player. Then, while playing, it rotates in place a certain amount per frame.

M. Dialogue Triggers

The primary focus of the dialogue triggers was discovering how to access and modify the grab interactable component in our scripts. We first developed a simple audio-trigger script that used a boolean to indicate if the dialogue for the corresponding element had been played or not.

However, in the development process, we realized we needed a way to manage multiple objects of the same type. So, we created a multi-trigger script that takes advantage of static booleans to track if a particular dialogue had been triggered yet by any one of its components. There is only one multitrigger audioscript For example, for type crayon, the first crayon that is picked up will play the dialogue. Then, on subsequent pickups of any crayon, the dialogue will not trigger again. We classified the different types of objects with multiple potential triggers with an enum. Each instance of that object had its associated enum:

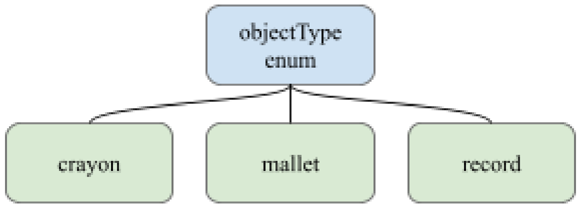

Fig. 6. Enum diagram for our multi-object type classification

N. Controls and Movement

The primary difficulty with technology and implementation at the start of this project was synching the controllers to player movement. After configuring the Quest 2 and switching from Unity’s 3D template to its VR template, we decided that the bloating from their VR template was worth the tradeoff of faster development time.

Because we were using Unity’s XR interaction toolkit, our primary challenges involved understanding and acquainting ourselves with the library.

The initial template for controls had a wide range of available motions: rotation, teleport movement, smooth movement, snap turning, tunnel vision, jumping, and grabbing. To maintain a minimalist control scheme and preserve accessibility, we opted for a smaller range of controls. We disabled snap turning, tunnel vision, jumping, and teleport movement as they were disorienting and did not suit the scale of our scene. We preserved smooth turning with the left joystick and rotation with the right joystick so that the game could be playable while seated. The bottom triggers on both controllers are responsible for grabbing items. We updated the UI to reflect these changes.

V. Analysis and Verification

By the end of Spring Quarter, we created a demoable, guided sandbox VR experience that provided a calming and creative environment.

Throughout our playtests, people saw the value in an experience like this. Those with family members with dementia provided valuable feedback on more intuitive controls and design choices. Because of this feedback, we adjusted movement speeds, added rotational movement, removed tunnelling, and added visual prompts for controls. We also included more items from the time period, like posters from older Disney movies. In an earlier round of playtesting, 5 out of 5 responders indicated that the perceived time period was 1990s or later. The final playtest showed that 9 out of 14 responders indicated that the perceived time period was between 1950s-1970s, which was our target.

Because our scripts are modular, the project should scale well. It is our intent to continue working on this project through the summer. We know that there are some existing faults discovered in our final playtest that were not resolved yet, for example, if the player moves their head too far outside of the room, they end up “falling out” of the window. We would also like to refine some of the colliders and their properties for more intuitive interactions.

Across the four rounds of playtesting, across the board, players agreed that controls became increasingly more intuitive. The perception of the time period became more accurate after adding in background music. Players were eager to explore the room, loved throwing things around, and were excited to push the limits of what they could do. By our final demo, we added in dialogue for a guided experience, and the consensus was that it successfully created a homey and comforting environment.

VI. Future Work

While the current version of Homebound meets its core design goals, several enhancements and expansions are planned to increase interactivity, improve polish, and deepen the emotional resonance of the experience.

O. Expanded Interactivity

Additional interactive elements are being considered to increase player agency and memory stimulation. These include functional drawers, train animation, and readable children's books. A journal system is also under development, allowing users to track which items they’ve interacted with or “remembered,” potentially creating a loose narrative structure.

P. Audio Refinement

While the maternal voiceover was effective in creating emotional comfort, future iterations will improve the audio layering system to better balance environmental ambiance with spoken dialogue. Specific sound design issues, such as the record scratch being overwritten by music playback, will be resolved by implementing timed audio triggers and adjustable voice priorities.

Q. Lighting and Visual Fidelity

More fine-grain control over lighting and shadows is planned to further enhance clarity and depth perception without overwhelming users. Additional art assets may also be introduced to support broader representation of childhood experiences across different backgrounds, potentially including multiple room layout options.

R. Expanded Spaces

A future version of *Homebound* may introduce new rooms or environments, such as a living room or outdoor scene. This would allow for a more complete memory journey and provide additional interaction variety. A short introductory sequence may also be added to ease players into the emotional tone of the experience before entering the childhood bedroom.

S. Interaction Smoothing and Accessibility

Planned updates to the grab system include support for multiple grip points and custom attach transforms for different objects. This will ensure more consistent and intuitive object handling across all item types. Additional accessibility options—such as adjustable text size, voice pitch, and movement sensitivity—will be added to support a wider range of user needs.

VII. Conclusion

Throughout the last two quarters, our group has worked on not only strengthening our knowledge of specific software and programs but also pushing one another past our normal limits to create a game that is functional and entertaining for the audience. In order to accomplish this goal to the best of our ability, we split our roles according to strengths and interests. With this in mind, Claire worked on configuring version control, Unity environment, programming scripts, and scene assembly, Ainsley worked on sourcing audio, creating models through Maya, and scene construction, and Emily worked on models in Blender, writing a script, and researching issues that were slowing down the workflow.

While the flowchart had changed halfway through the second quarter due to time restraints and certain aspects taking longer than originally expected, our team made the best of what we had and worked on exponentially growing the experience itself. With this in mind, we focused on user feedback and by the end of the quarter had a game that allowed people to laugh, smile, and feel overall satisfied with our game, “Homebound”. We reduced motion blur, delved into research of the period to get an appropriate scope of what our designated audience would see as familiar, and implemented that research through appropriate models such as a wooden xylophone, Crayons with the proper logo, 50’s 50s-based records, and so forth. Overall, this is a project all three of us are proud of, not only for developing it, but for pushing one another to not settle for anything less than what we imagined initially.

Final Demo

Our final demo is linked here: https://youtu.be/gZWxDhkA5ls?si=nilPjgF8pWNoPUlo

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